

DSCI II Final Project

Heather Lu

2026-05-11

Load Dataset + Clean

```
# Load the full dataset
load("/Users/heatherlu/Desktop/DSII_final/severity.RData")

# Set the random seed to the last 4 digits of YOUR UNI
set.seed(3978)

# Randomly sample 1,000 participants
my_dat <- dat[sample(nrow(dat), 1000), ]

# Convert categorical variables to factors
my_dat <- my_dat |>
  mutate(
    severity = factor(severity, levels = c(1, 0), labels = c("severe", "not_severe")),
    gender = factor(gender),
    race = factor(race),
    smoking = factor(smoking),
    hypertension = factor(hypertension),
    diabetes = factor(diabetes),
    vaccine = factor(vaccine)
  ) |>
  dplyr::select(-id)

# 80/20 train/test split
set.seed(3978)
data_split <- initial_split(my_dat, prop = 0.8)
training_data <- training(data_split)
testing_data <- testing(data_split)
```

EDA

```
# Summary statistics table by severity group
summary_table <- my_dat |>
  group_by(severity) |>
  summarise(
    n = n(),
```

```

age_mean_sd = paste0(round(mean(age), 1), " (", round(sd(age), 1), ")"),
bmi_mean_sd = paste0(round(mean(bmi), 1), " (", round(sd(bmi), 1), ")"),
sbp_mean_sd = paste0(round(mean(SBP), 1), " (", round(sd(SBP), 1), ")"),
ldl_mean_sd = paste0(round(mean(LDL), 1), " (", round(sd(LDL), 1), ")"),
dep_mean_sd = paste0(round(mean(depression), 1), " (", round(sd(depression), 1), ")"),
pct_male = paste0(round(mean(gender == 1) * 100, 1), "%"),
pct_vaccinated = paste0(round(mean(vaccine == 1) * 100, 1), "%"),
pct_htn = paste0(round(mean(hypertension == 1) * 100, 1), "%"),
pct_diabetes = paste0(round(mean(diabetes == 1) * 100, 1), "%")
) |>
t() |>
as.data.frame()

kable(summary_table,
  col.names = c("Severe", "Not Severe"),
  caption = "Summary Statistics by severity group.")

```

Table 1: Summary Statistics by severity group.

	Severe	Not Severe
severity	severe	not_severe
n	321	679
age_mean_sd	61 (4.2)	59.5 (4.4)
bmi_mean_sd	28.2 (2.8)	27.4 (2.6)
sbp_mean_sd	132.6 (7.3)	128.5 (8)
ldl_mean_sd	111.2 (18.7)	109.3 (20.1)
dep_mean_sd	7 (1.8)	6.9 (2)
pct_male	49.8%	49.6%
pct_vaccinated	20.9%	80.4%
pct_htn	59.8%	38.7%
pct_diabetes	13.1%	15.2%

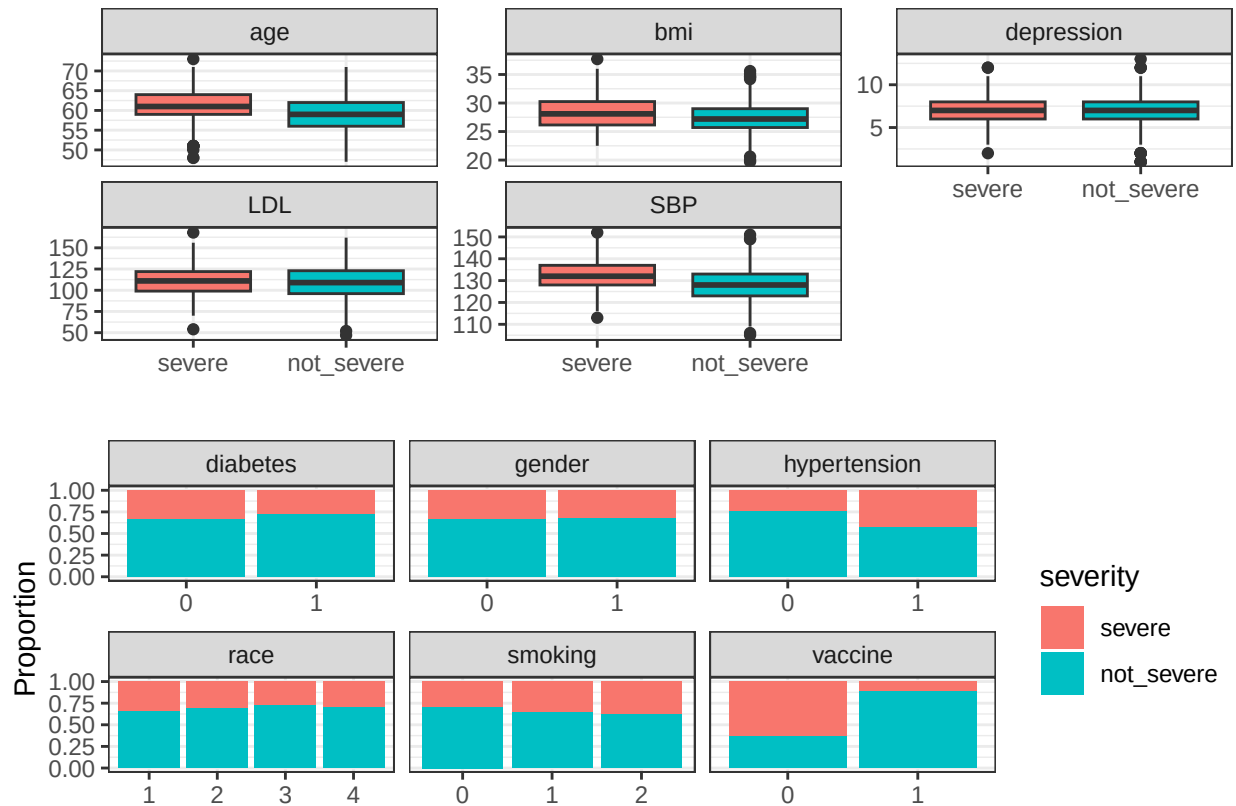
```

# Figure 1a: box plots for continuous predictors
p1 <- training_data |>
  dplyr::select(severity, age, bmi, SBP, LDL, depression) |>
  pivot_longer(cols = -severity, names_to = "variable", values_to = "value") |>
  ggplot(aes(x = severity, y = value, fill = severity)) +
  geom_boxplot() +
  facet_wrap(~ variable, scales = "free_y") +
  labs(x = "", y = "") +
  theme_bw() +
  theme(legend.position = "none")

# Figure 1b: bar chart for categorical predictors
p2 <- training_data |>
  dplyr::select(severity, gender, race, smoking, hypertension, diabetes, vaccine) |>
  pivot_longer(cols = -severity, names_to = "variable", values_to = "value") |>
  ggplot(aes(x = value, fill = severity)) +
  geom_bar(position = "fill") +
  facet_wrap(~ variable, scales = "free_x") +
  labs(y = "Proportion", x = "") +
  theme_bw()

```

```
# Combine
grid.arrange(p1, p2, nrow = 2)
```



Model Training

```
# ctrl object: shared across all 4 models
ctrl <- trainControl(method = "cv",
                     number = 10,
                     summaryFunction = twoClassSummary,
                     classProbs = TRUE)
```

Model 1: Logistic Regression

```
set.seed(3978)
model.glm <- train(severity ~ .,
                  data = training_data,
                  method = "glm",
                  metric = "ROC",
                  trControl = ctrl)

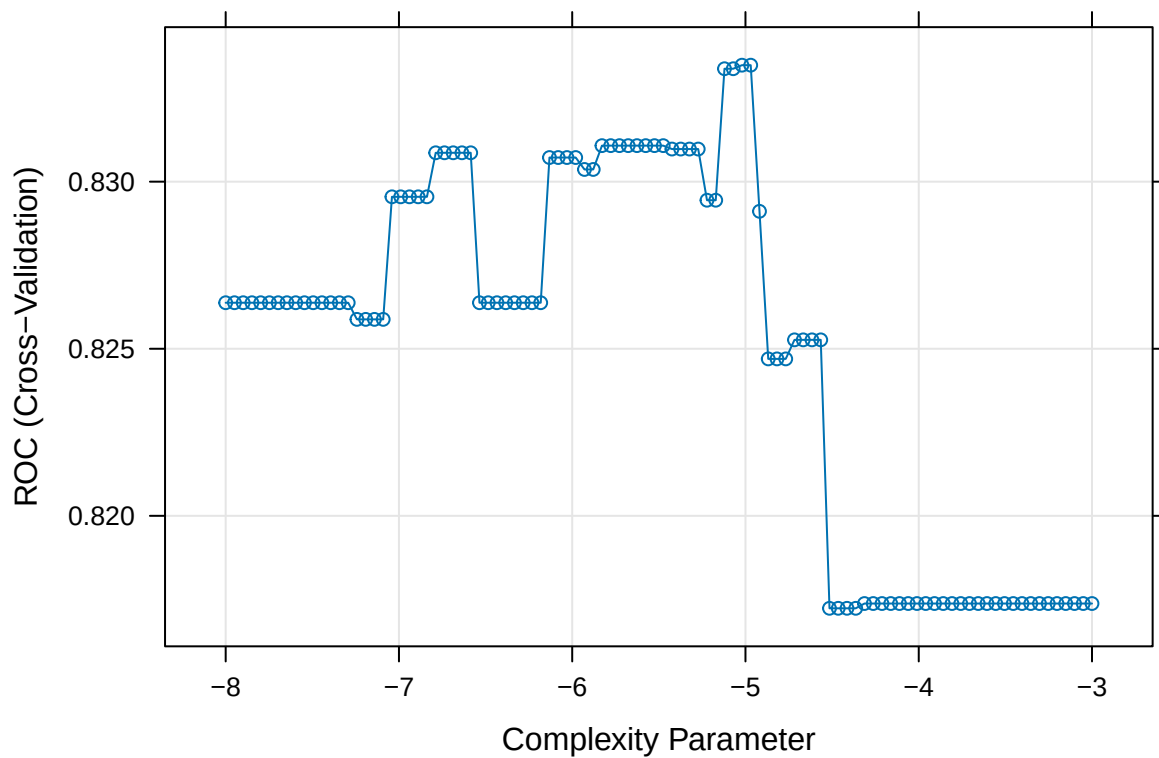
model.glm$results
```

##	parameter	ROC	Sens	Spec	ROCSD	SensSD	SpecSD
## 1	none	0.8608924	0.676	0.8854209	0.04401017	0.07617137	0.02575705

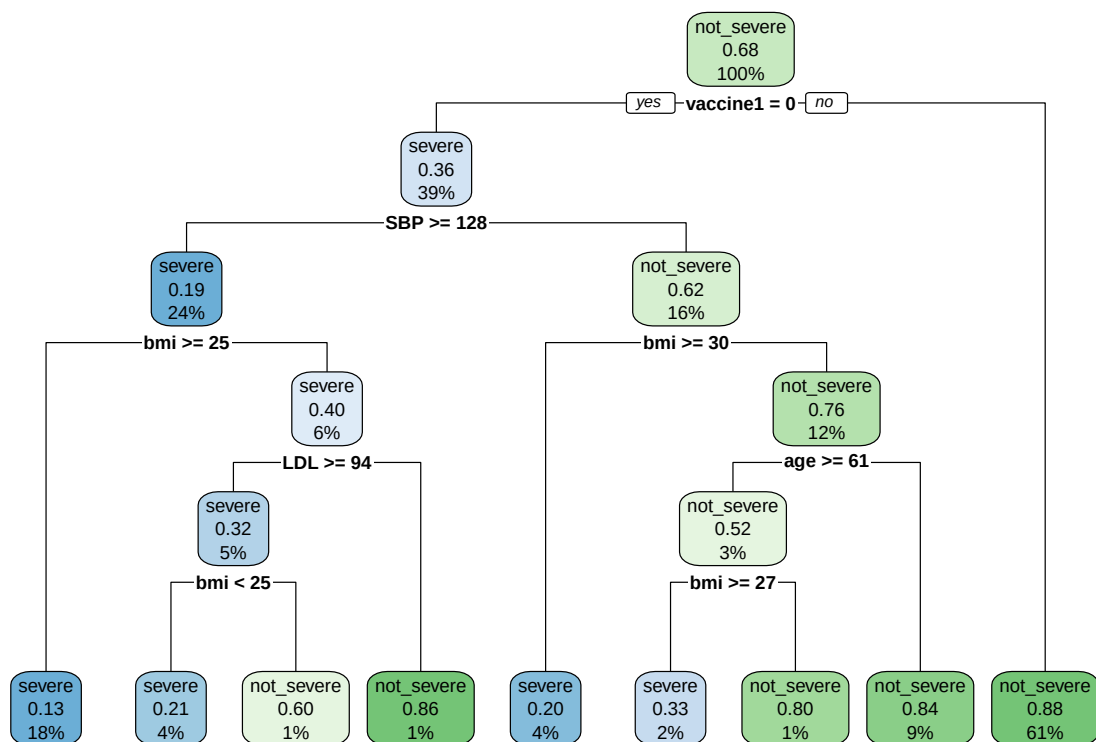
Model 2: Decision Tree

```
set.seed(3978)
model.rpart <- train(severity ~ .,
  data = training_data,
  method = "rpart",
  tuneGrid = data.frame(cp = exp(seq(-8, -3, len = 100))),
  metric = "ROC",
  trControl = ctrl)

# CV plot - ROC vs cp (log scale)
plot(model.rpart, xTrans = log)
```



```
# Plot the final pruned tree
rpart.plot(model.rpart$finalModel)
```



```
# Best cp value
model.rpart$bestTune
```

```
##           cp
## 61 0.006945252
```

Model 3: AdaBoost

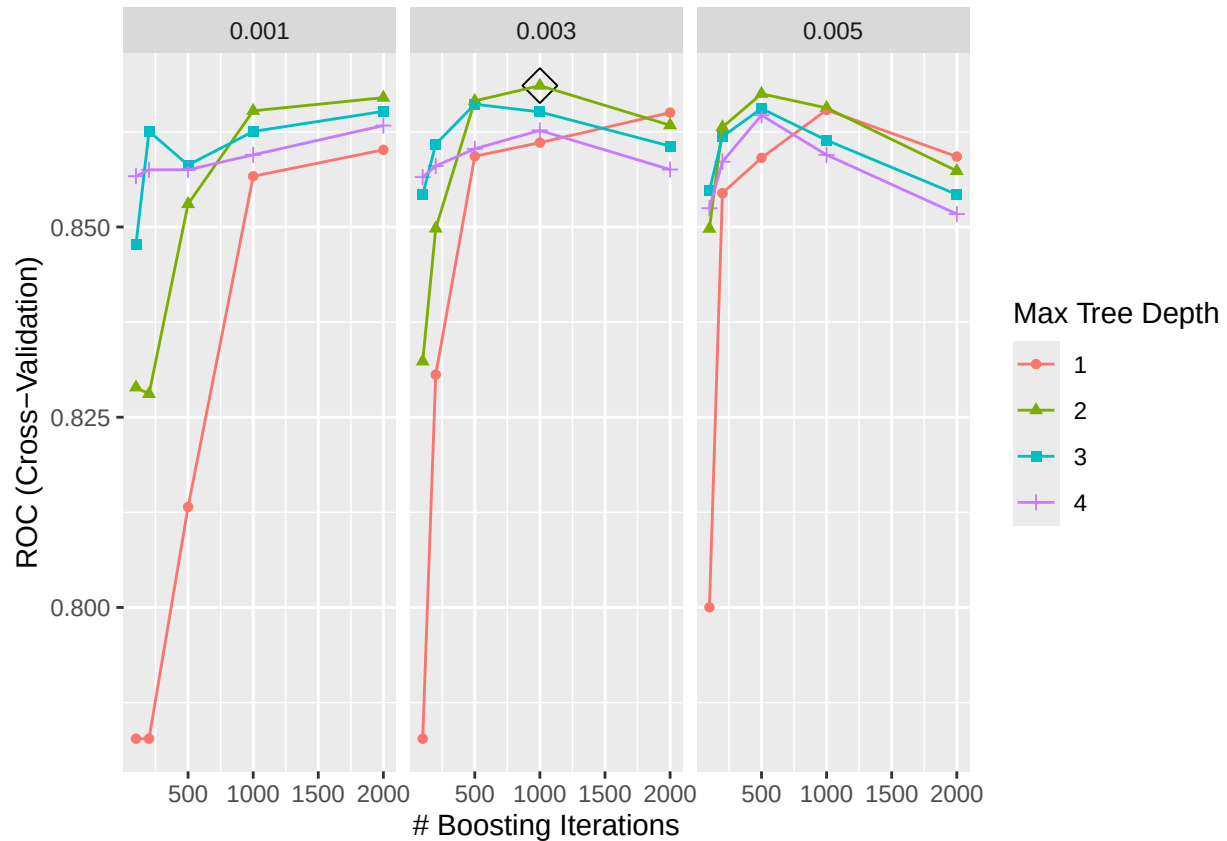
```
gbmA.grid <- expand.grid(n.trees = c(100, 200, 500, 1000, 2000),
  interaction.depth = 1:4,
  shrinkage = c(0.001, 0.003, 0.005),
  n.minobsinnode = 10)
```

```
set.seed(3978)
model.gbma <- train(severity ~ .,
  data = training_data,
  method = "gbm",
  distribution = "adaboost",
  tuneGrid = gbmA.grid,
  metric = "ROC",
  trControl = ctrl,
  verbose = FALSE)
```

```
model.gbma$bestTune
```

```
##      n.trees interaction.depth shrinkage n.minobsinnode
## 29      1000                2      0.003              10
```

```
ggplot(model.gbmA, highlight = TRUE)
```



Model 4: SVM-Radial

```
svmr.grid <- expand.grid(C = exp(seq(1, 7, len = 30)),
                        sigma = exp(seq(-10, -2, len = 15)))

set.seed(3978)
model.svmr <- train(severity ~ .,
                    data = training_data,
                    method = "svmRadialSigma",
                    tuneGrid = svmr.grid,
                    metric = "ROC",
                    trControl = ctrl)
```

```
## line search fails -0.8964073 -0.4411244 1.050201e-05 4.275942e-06 -6.310528e-08 -2.323497e-08 -7.620
```

```
## line search fails -0.5153714 -0.5467036 1.347303e-05 2.89005e-06 -3.062256e-08 -1.802433e-08 -4.6466
```

```
## line search fails -0.7488311 -0.3744341 1.020376e-05 2.840393e-06 -4.038179e-08 -8.977903e-09 -4.375
```

```
## line search fails -0.6256962 -0.4949248 1.141636e-05 3.516476e-06 -3.790244e-08 -2.105642e-08 -5.067
```

```
## line search fails -0.4527578 -0.5546846 1.124066e-05 2.444664e-06 -2.083522e-08 -1.630861e-08 -2.740
```

```
## line search fails -0.5518581 -0.5204091 1.062953e-05 2.671932e-06 -2.735643e-08 -1.626007e-08 -3.342
```

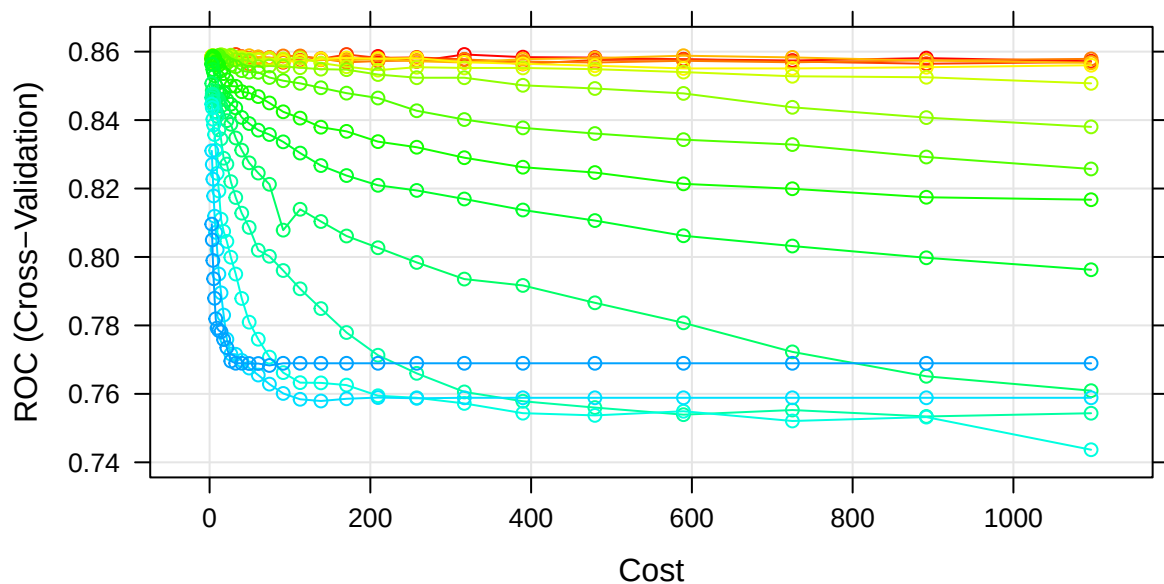
```
model.svmr$bestTune
```

```
##          sigma          C
## 346 4.539993e-05 316.9109
```

```
myCol <- rainbow(25)
myPar <- list(superpose.symbol = list(col = myCol),
             superpose.line   = list(col = myCol))

plot(model.svmr, highlight = TRUE, par.settings = myPar)
```

Sigma			
○ ○ ○	0.000446404210737614	○ ○ ○	0.00438936184277844
○ ○ ○	0.000790490323119966	○ ○ ○	0.00777265979560353
○ ○ ○	0.00139979627412967	○ ○ ○	0.0137637867330504
○ ○ ○	0.00247875217666636	○ ○ ○	0.0243728440732796



Model Comparison

```
resamp <- resamples(list(GLM = model.glm,
                        RPART = model.rpart,
```

```

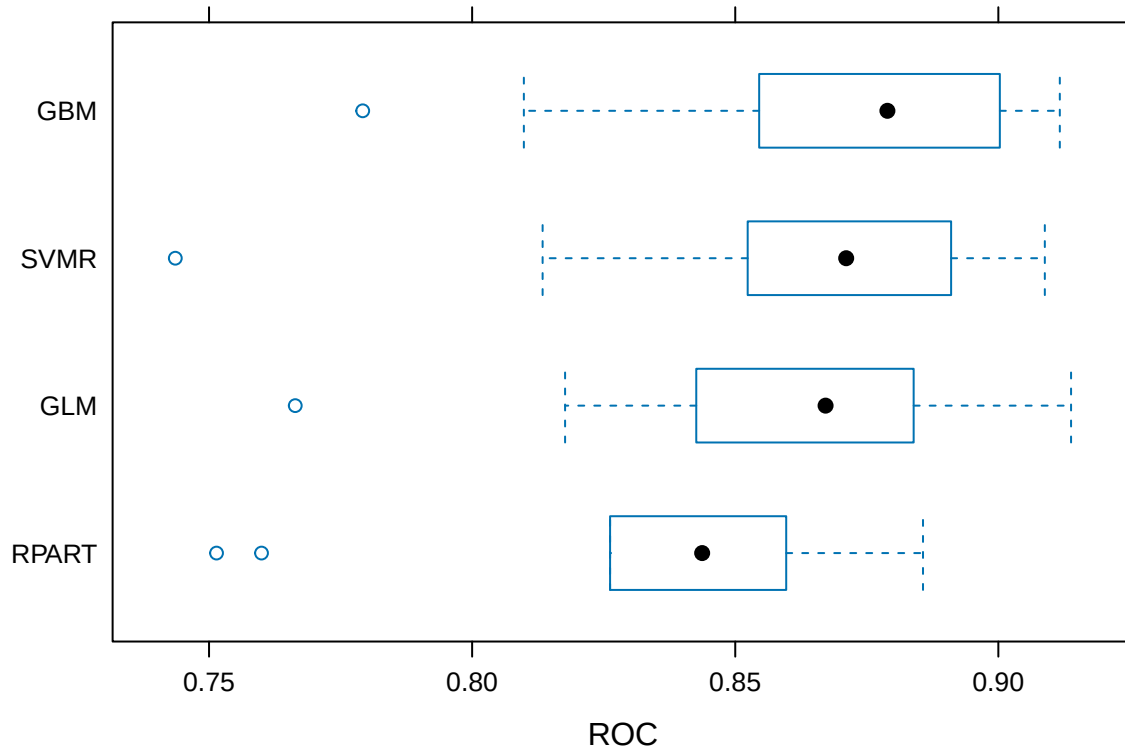
GBM = model.gbmA,
SVMR = model.svmr))

summary(resamp)

##
## Call:
## summary.resamples(object = resamp)
##
## Models: GLM, RPART, GBM, SVMR
## Number of resamples: 10
##
## ROC
##      Min.    1st Qu.    Median      Mean   3rd Qu.      Max. NA's
## GLM    0.7663818 0.8468535 0.8671652 0.8608924 0.8831909 0.9138177    0
## RPART 0.7514245 0.8267450 0.8437141 0.8334864 0.8589744 0.8856838    0
## GBM    0.7792023 0.8602823 0.8789174 0.8685742 0.8985043 0.9116809    0
## SVMR   0.7435897 0.8533041 0.8710826 0.8591680 0.8901353 0.9088319    0
##
## Sens
##      Min.    1st Qu.    Median      Mean   3rd Qu.      Max. NA's
## GLM    0.5769231 0.6538462 0.6538462 0.6760000 0.6826923 0.8461538    0
## RPART 0.5000000 0.6538462 0.6730769 0.6793846 0.6923077 0.8461538    0
## GBM    0.5000000 0.6250000 0.6538462 0.6603077 0.6734615 0.8461538    0
## SVMR   0.5769231 0.6923077 0.7307692 0.7338462 0.7923077 0.8461538    0
##
## Spec
##      Min.    1st Qu.    Median      Mean   3rd Qu.      Max. NA's
## GLM    0.8518519 0.8703704 0.8808081 0.8854209 0.9074074 0.9259259    0
## RPART 0.7777778 0.8703704 0.8981481 0.8928620 0.9259259 0.9629630    0
## GBM    0.8727273 0.8935185 0.9166667 0.9224579 0.9444444 0.9814815    0
## SVMR   0.6296296 0.7535354 0.8055556 0.8097306 0.8842593 0.9259259    0

# Box plot of CV ROC across models
bwplot(resamp, metric = "ROC")

```



```
# Table 2: CV performance summary
cv_summary <- data.frame(
  Model = c("Logistic Regression", "Decision Tree", "AdaBoost", "SVM-Radial"),
  Mean_AUC = c(0.861, 0.833, 0.868, 0.859),
  Median_AUC = c(0.867, 0.844, 0.861, 0.871),
  Mean_Sensitivity = c(0.676, 0.679, 0.660, 0.734),
  Mean_Specificity = c(0.885, 0.893, 0.915, 0.810)
)

kable(cv_summary,
  caption = "Cross-validated model performance across 10 folds.",
  col.names = c("Model", "Mean AUC", "Median AUC",
    "Mean Sensitivity", "Mean Specificity"),
  digits = 3)
```

Table 2: Cross-validated model performance across 10 folds.

Model	Mean AUC	Median AUC	Mean Sensitivity	Mean Specificity
Logistic Regression	0.861	0.867	0.676	0.885
Decision Tree	0.833	0.844	0.679	0.893
AdaBoost	0.868	0.861	0.660	0.915
SVM-Radial	0.859	0.871	0.734	0.810

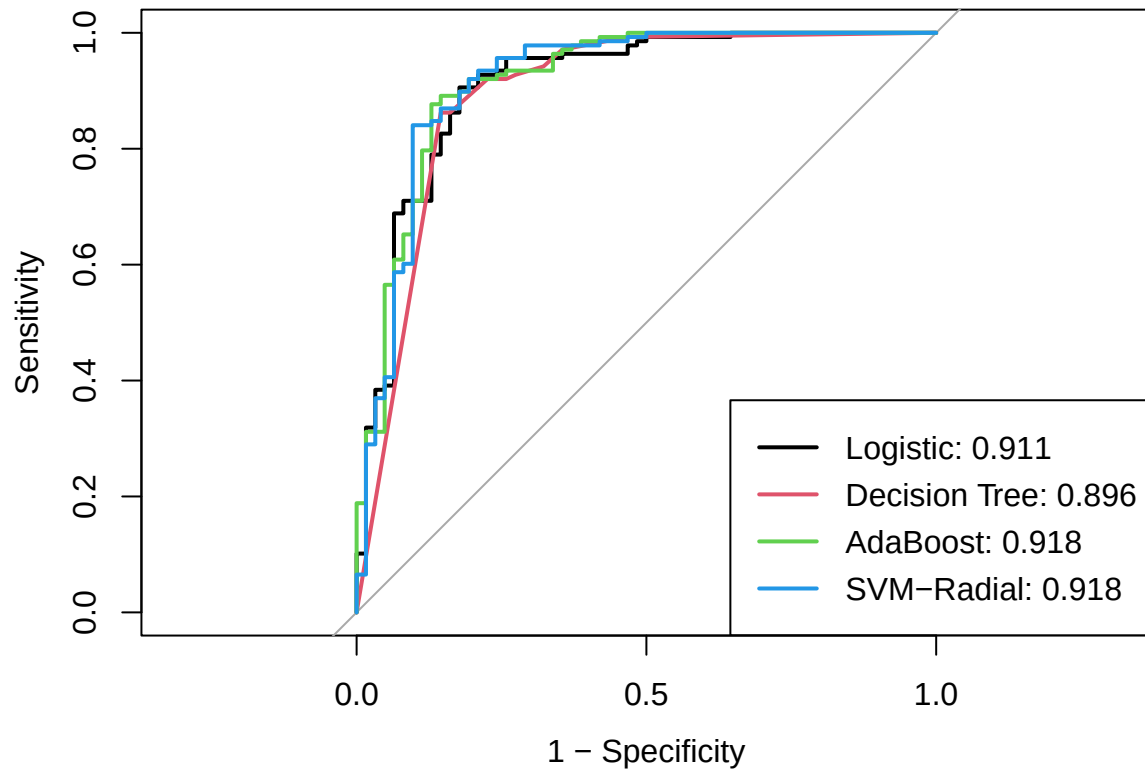
Test set performance & ROC curves

```
# Get predicted probabilities on test set for all 4 models
glm.pred <- predict(model.glm, newdata = testing_data, type = "prob")[, "severe"]
rpart.pred <- predict(model.rpart, newdata = testing_data, type = "prob")[, "severe"]
gbmA.pred <- predict(model.gbmA, newdata = testing_data, type = "prob")[, "severe"]
svmr.pred <- predict(model.svmr, newdata = testing_data, type = "prob")[, "severe"]

# Compute ROC curves
roc.glm <- roc(testing_data$severity, glm.pred, quiet = TRUE)
roc.rpart <- roc(testing_data$severity, rpart.pred, quiet = TRUE)
roc.gbmA <- roc(testing_data$severity, gbmA.pred, quiet = TRUE)
roc.svmr <- roc(testing_data$severity, svmr.pred, quiet = TRUE)

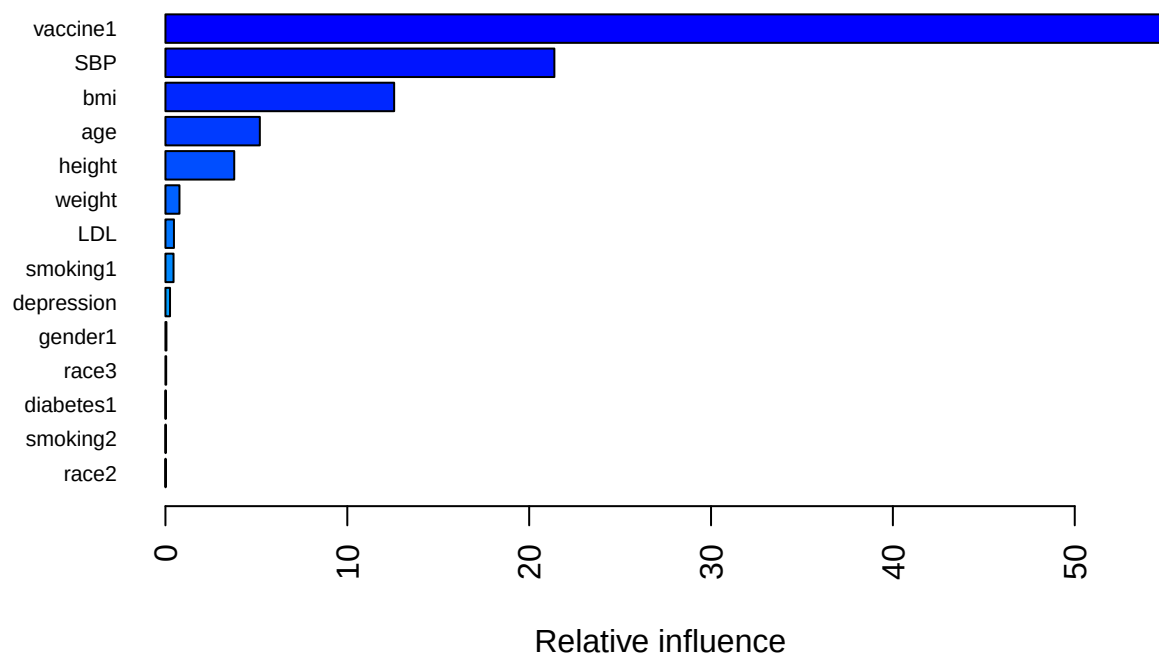
# Plot all ROC curves together
plot(roc.glm, legacy.axes = TRUE, col = 1)
plot(roc.rpart, add = TRUE, col = 2)
plot(roc.gbmA, add = TRUE, col = 3)
plot(roc.svmr, add = TRUE, col = 4)

# AUC values in legend
auc_vals <- c(roc.glm$auc[1], roc.rpart$auc[1],
              roc.gbmA$auc[1], roc.svmr$auc[1])
modelNames <- c("Logistic", "Decision Tree", "AdaBoost", "SVM-Radial")
legend("bottomright",
      legend = paste0(modelNames, ": ", round(auc_vals, 3)),
      col = 1:4, lwd = 2)
```



Variable Importance

```
# Variable importance from best model (AdaBoost)
summary(model.gbmA$finalModel,
  las = 2,
  cBars = 14,
  cex.names = 0.7)
```



```
##           var      rel.inf
## vaccine1    vaccine1 54.98210082
## SBP          SBP    21.38670098
## bmi          bmi    12.57688475
## age          age     5.19026938
## height      height   3.78292658
## weight      weight   0.77120979
## LDL         LDL     0.46772565
## smoking1    smoking1 0.44548508
## depression  depression 0.25288367
## gender1     gender1  0.05383525
## race3       race3    0.04169065
## diabetes1   diabetes1 0.01794765
## smoking2    smoking2 0.01748147
## race2       race2    0.01285829
## race4       race4    0.00000000
## hypertension1 hypertension1 0.00000000
```

PDP for top 3 variables

```
# PDP for vaccine
pdp.vaccine <- model.gbmA |>
  pdp::partial(pred.var = "vaccine",
    grid.resolution = 100,
```

```

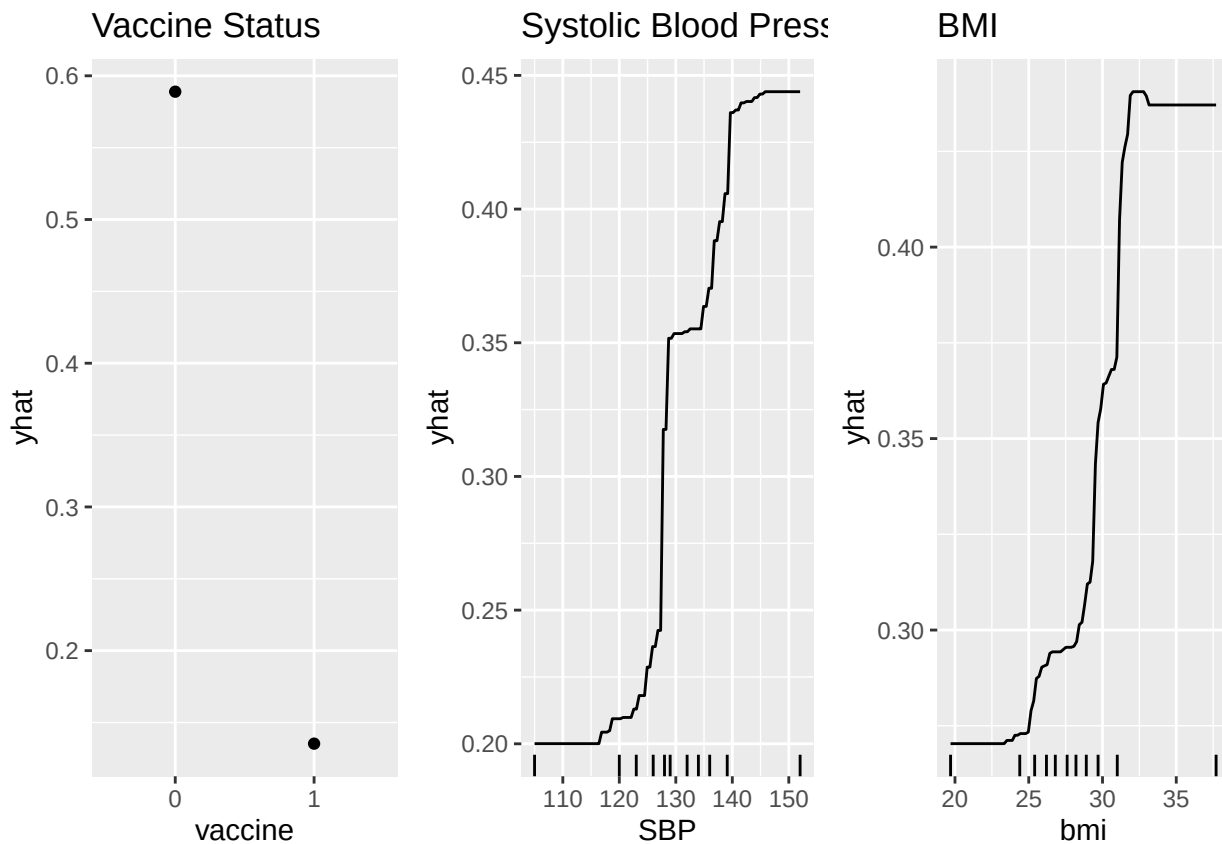
        prob = TRUE) |>
autoplot(rug = TRUE, train = training_data) +
ggtitle("Vaccine Status")

# PDP for SBP
pdp.sbp <- model.gbmA |>
  pdp::partial(pred.var = "SBP",
    grid.resolution = 100,
    prob = TRUE) |>
  autoplot(rug = TRUE, train = training_data) +
  ggtitle("Systolic Blood Pressure")

# PDP for BMI
pdp.bmi <- model.gbmA |>
  pdp::partial(pred.var = "bmi",
    grid.resolution = 100,
    prob = TRUE) |>
  autoplot(rug = TRUE, train = training_data) +
  ggtitle("BMI")

# Combine into one figure
gridExtra::grid.arrange(pdp.vaccine, pdp.sbp, pdp.bmi, nrow = 1)

```



```

# Predicted risk scores from AdaBoost on test set
risk_scores <- data.frame(

```

```

predicted_prob = gbmA.pred,
true_severity = testing_data$severity
) |>
  arrange(desc(predicted_prob))

# Top 10 highest risk patients
risk_scores |>
  head(10) |>
  kable(caption = "Top 10 patients with highest predicted risk of severe COVID-19 infection.",
        col.names = c("Predicted Probability", "True Severity"),
        digits = 3)

```

Table 3: Top 10 patients with highest predicted risk of severe COVID-19 infection.

Predicted Probability	True Severity
0.904	severe
0.892	severe
0.885	severe
0.874	severe
0.867	severe
0.857	severe
0.856	severe
0.820	severe
0.819	severe
0.813	severe

```

# Distribution of risk scores by true severity group
risk_scores |>
  group_by(true_severity) |>
  summarise(
    mean_risk = round(mean(predicted_prob), 3),
    median_risk = round(median(predicted_prob), 3),
    min_risk = round(min(predicted_prob), 3),
    max_risk = round(max(predicted_prob), 3)
  ) |>
  kable(caption = "Distribution of predicted risk scores by true severity group.",
        col.names = c("Severity Group", "Mean", "Median", "Min", "Max"))

```

Table 4: Distribution of predicted risk scores by true severity group.

Severity Group	Mean	Median	Min	Max
severe	0.602	0.700	0.085	0.904
not_severe	0.184	0.141	0.061	0.692